

11-711 Homework 2: Retrieval-Augmented Generation for Pittsburgh and CMU

Your Name

yourandrewid@andrew.cmu.edu

Due: October 23, 2026, 11:59 PM ET

Abstract

rich Küttler, Mike Lewis, Wen tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2021. [Retrieval-augmented generation for knowledge-intensive nlp tasks](#). *Preprint*, arXiv:2005.11401.

1 Introduction

We build a retrieval-augmented generation system (Lewis et al., 2021) that answers factual questions about Pittsburgh and Carnegie Mellon University.

2 Data Creation

2.1 Knowledge resource

2.2 Extraction and cleaning

2.3 Annotated training data

2.4 Other data

3 Model Details

3.1 Chunking

3.2 Dense retrieval

3.3 Sparse retrieval

3.4 Hybrid retrieval

3.5 Reader

3.6 Baselines

4 Results

5 Analysis

5.1 Performance by question type

5.2 Comparing your variations

5.3 Retrieval versus closed-book

5.4 Dense, sparse, and hybrid

5.5 Example outputs

6 Conclusion

Limitations

Use of AI (required, ungraded)

References

Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Hein-